⚙️ Important DSA Algorithms and Their Uses

🧩 **Algorithm Name**	📘 **Category**	⚡ **Use / Purpose**
Dijkstra's Algorithm	Graph	Finds the shortest path from a source to all other nodes (no negative weights).
Bellman-Ford Algorithm	Graph	Shortest path algorithm that works with negative weights.
Floyd-Warshall Algorithm	Graph	Finds shortest paths between all pairs of vertices.
A* (A-Star) **Algorithm**	Graph / AI	Pathfinding and graph traversal using heuristics (used in games, maps).
Kruskal's Algorithm	Graph	Finds Minimum Spanning Tree (MST) — connects all nodes with minimal cost.
Prim's Algorithm	Graph	Another MST algorithm — grows MST from a starting vertex.
Depth First Search (DFS)	Graph / Tree	Explores as far as possible along each branch before backtracking.
Breadth First Search (BFS)	Graph / Tree	Explores all neighbors level by level (used in shortest unweighted paths).
Binary Search	Searching	Efficiently searches for an element in a sorted array (O(log n)).
Merge Sort	Sorting	Stable, divide-and-conquer sorting algorithm (O(n log n)).
Quick Sort	Sorting	Fast in practice, uses partitioning (O(n log n) average).
Heap Sort	Sorting	Uses heap data structure for in-place sorting.
Dynamic Programming (DP)	Optimization	Breaks problems into overlapping subproblems for optimization.
Kadane's Algorithm	Dynamic Programming	Finds maximum subarray sum efficiently.
KMP Algorithm	String	Efficient pattern matching (substring search).
Rabin-Karp Algorithm	String	Fast string search using hashing.
Topological Sort	Graph	Orders vertices in a Directed Acyclic Graph (DAG).
Union-Find (Disjoint Set Union)	Graph	Detects cycles and helps in Kruskal's MST algorithm.
Tarjan's Algorithm	Graph	Finds strongly connected components (SCCs).

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Backtracking (e.g., N-Queens, Sudoku)	Recursion	Tries all possibilities recursively to find valid solutions.
Greedy Algorithms	Optimization	Makes locally optimal choices (e.g., Huffman coding, activity selection).